

Route decision: Mumbai JNPT Logistics Park → Chennai Logistics Park

HUB-MUMBAI → HUB-CHENNAI · mode any (STRICT) · cargo general · priority normal · scenarios: CHENNAI_FLOOD · planned 2026-09-29 13:47 UTC

Recommendation: SAFEST / BALANCED, ROAD via direct. Expected arrival in **76 h** (3.2 days); plan for 78 h (p85), worst case 97 h (p95). Cost: freight INR 187,148. Peak leg risk 75%.

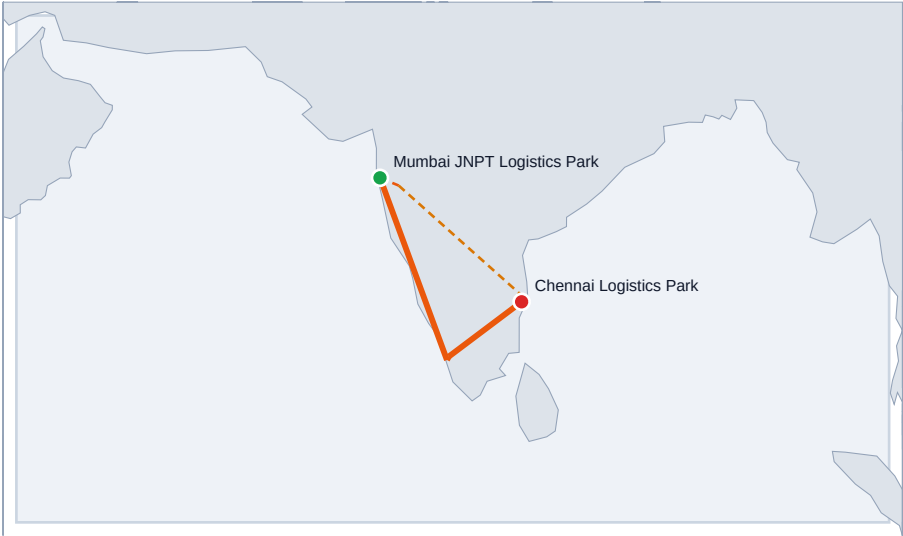
Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Risk
FASTEST	AIR	direct	63	70	75	92	INR 1,359,300	75%
SAFEST / BALANCED	ROAD	direct	71	76	78	97	INR 187,148	75%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 70.4 h (p85 74.6 h, p95 91.8 h), INR 1,359,300 freight, 2 handoffs. 5.9 h faster than the next option. INR 1,172,149 more (freight) than SAFEST. Includes 48.0 h announced scenario delay.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Pune Air Cargo Hub	106	1.6	+3.6	INR 12,183	0%	
2	handoff	Pune Air Cargo Hub		6.0	+0.0	INR 14,410	2%	
3	AIR	Chennai International Airport	912	1.1	+8.8	INR 1,314,979	0%	
4	handoff	Chennai International Airport		6.0	+0.0	INR 14,410	2%	
5	ROAD	Chennai Logistics Park	26	48.4	+0.0	INR 3,017	75%	SCENARIO: Severe urban flooding. Inland road access to Port and Logistics Park is underwater.

Transit 3.1 h + transfers 12.0 h + announced scenario delay 48.0 h = plan 63.1 h. Model delay buffer p50/p85/p95: +7.4 / +12.5 / +36.2 h. Route band p50 70.4 h, p85 74.6 h, p95 91.8 h (independent legs); 99.3 h if all legs are late together. Costs: linehaul INR 1,330,178, transfers INR 28,821, disruption surcharge INR 302.

Why the slowest leg (AIR-PUNE → AIR-CHENNAI, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST / BALANCED: ROAD via direct

ROAD via no chokepoints; expected 76.3 h (p85 77.8 h, p95 96.8 h), INR 187,148 freight, 0 handoffs. 5.9 h slower than FASTEST. INR 1,172,149 cheaper (freight) than the lowest-cost alternative. Includes 48.0 h announced scenario delay.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Kochi Logistics Distribution Center	1,060	15.6	+3.6	INR 122,168	0%	
2	ROAD	Cochin International Airport	22	0.3	+3.6	INR 2,492	0%	
3	ROAD	Chennai Logistics Park	493	55.2	+0.0	INR 56,808	75%	SCENARIO: Severe urban flooding. Inland road access to Port and Logistics Park is underwater.

Transit 23.1 h + transfers 0.0 h + announced scenario delay 48.0 h = plan 71.1 h. Model delay buffer p50/p85/p95: +5.2 / +7.3 / +29.6 h. Route band p50 76.3 h, p85 77.8 h, p95 96.8 h (independent legs); 100.7 h if all legs are late together. Costs: linehaul INR 181,467, transfers INR 0, disruption surcharge INR 5,681.

Why the slowest leg (HUB-MUMBAI → DC-KOCHI, p85 3.6 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.